DATA ANALYSIS PORTFOLIO | UN SDG 6: CLEAN WATER AND SANITATION

Who still lacks safe drinking water, and is it improving?

An end-to-end analysis of WHO/UNICEF JMP estimates: 213 countries in 2020, and 231 countries tracked from 2015 to 2020.

Joshua | Google Sheets + Python

PROJECT OVERVIEW

A two-part study of access to drinking water

Why it matters

UN SDG 6 calls for safe, affordable drinking water for everyone. The WHO/UNICEF Joint Monitoring Programme ranks service on a ladder: at least basic, limited, unimproved, surface water.

Questions I set out to answer

- Who lacks access: by area, population size, income?
- Is access improving, and how fast?
- Which regions are furthest behind?

213

countries in the 2020 snapshot (Part 1)

231

countries tracked 2015 to 2020 (Part 2)

12

access measures: 4 service levels x national, rural, urban

+19

engineered features on top of the 16 original columns

Four findings: a rural gap, an income gap, uneven progress, one region behind

~768M

people in the 2020 data lack at least basic drinking water (about 1 in 10)

13.4 pts

rural-urban gap in basic access: 81.3% rural vs 94.7% urban (country means)

63% vs 99.6%

basic access in low-income vs high-income countries: income is the sharpest divider

135 of 231

countries improved national access 2015-2020; 62 were already at full access, 16 stalled, 16 declined

Sub-Saharan Africa has the fastest national gains (0.56 pp a year) yet the furthest to go: 37 of the 44 countries below 80% basic access are there.

APPROACH

From messy imports to measured insight in four stages

1

Import and clean

- Fix semicolon-split rows
- Handle NAN and null markers
- Cap impossible values >100%

2

Engineer features

- Urban and rural counts
- Year gap, ARC columns
- Full-access flags, region

3

Summarise

- Mean, median, mode, IQR, SD
- Pivot by income group
- ARC averages and counts

4

Visualise

- Box plots, stacked columns
- Histograms, bubble chart
- Plain-language takeaways

Spreadsheet toolkit COUNTA, IF / IFERROR, LOOKUP, QUARTILE, ROUNDUP, COUNTIFS, pivot tables, native charts
Python reproduction pandas and matplotlib re-derive every metric; 84 of 84 checks match the workbooks

Cleaning came first: three import problems fixed

Semicolon-split rows

5 rows imported into a single cell. Added value_cnt = COUNTA(row), filtered out the expected 16, then used Split text to columns on only the affected cells.

Shares above 100%

9 basic-access values exceeded 100% (max 100.0000037). Created wat_bas_n (rounded), so no share exceeds 100%.

Missing values

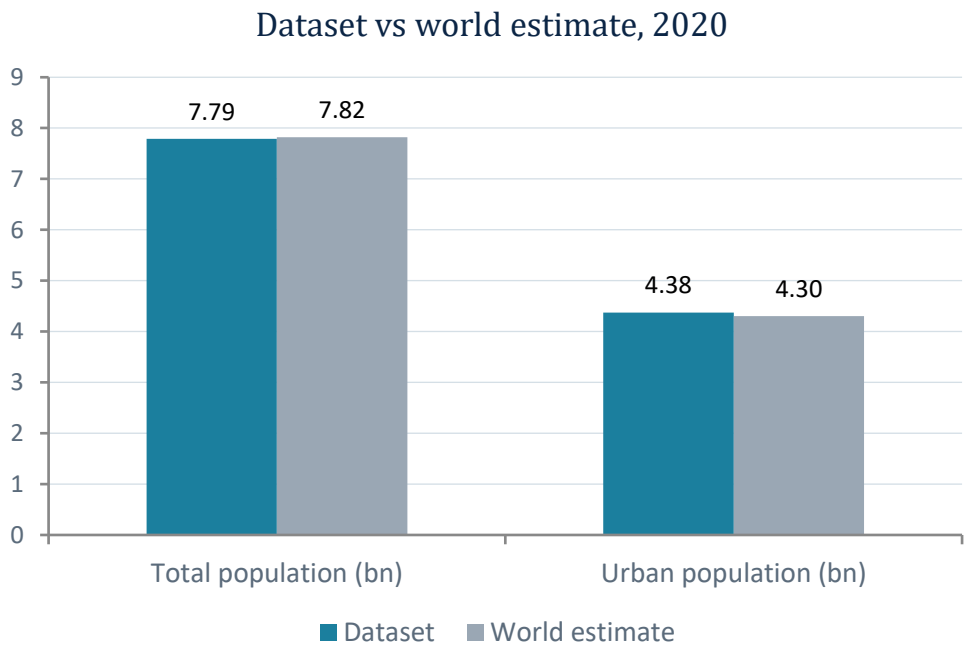
Missing estimates arrive as text (NAN). Replaced #VALUE! rounding errors with NAN so MAX, AVERAGE and QUARTILE ignore them.

The check that caught them

```
value_cnt
= COUNTA(A2:P2)
```

Every clean row returns 16. In the final data all 213 country rows do.

The dataset covers 99.6% of the world's people



0.44%

difference in total population (dataset vs 7.821 bn world)

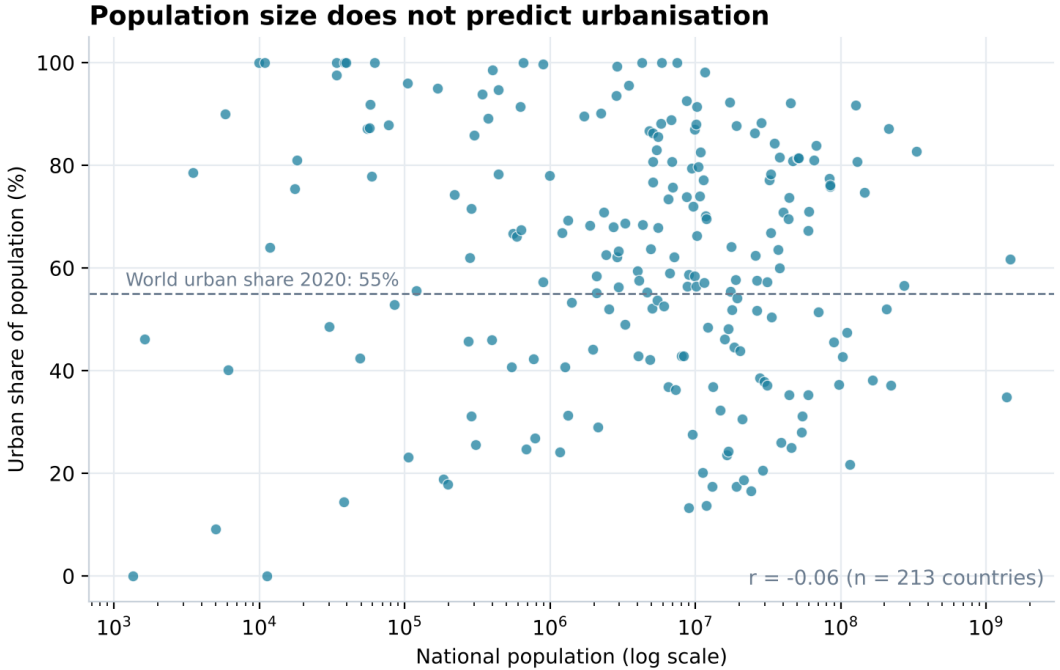
56.2%

urban share in the dataset vs 55% for the world

1.70%

difference in urban head count vs the world estimate

Population size says nothing about how urban a country is



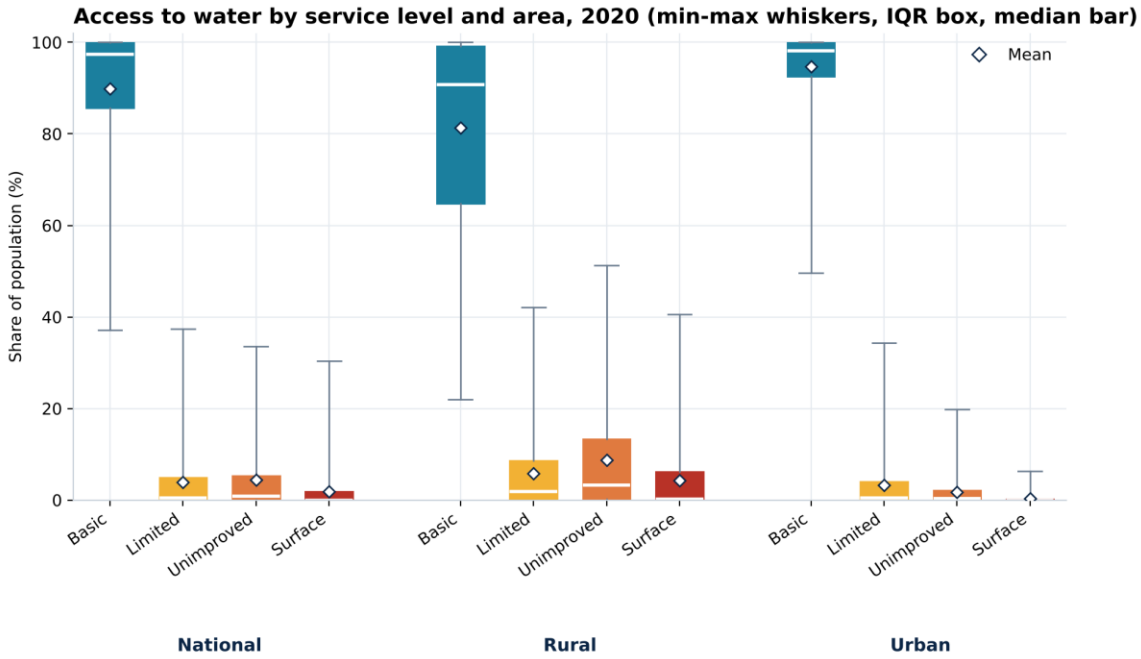
r = -0.06

Large and small countries are equally likely to be mostly urban or mostly rural.

Chart decision log

- Outliers (China, India) crushed the first line chart
- Chose a unit change over deleting data
- A log axis keeps all 213 countries visible

Rural access is both lower and far less even than urban

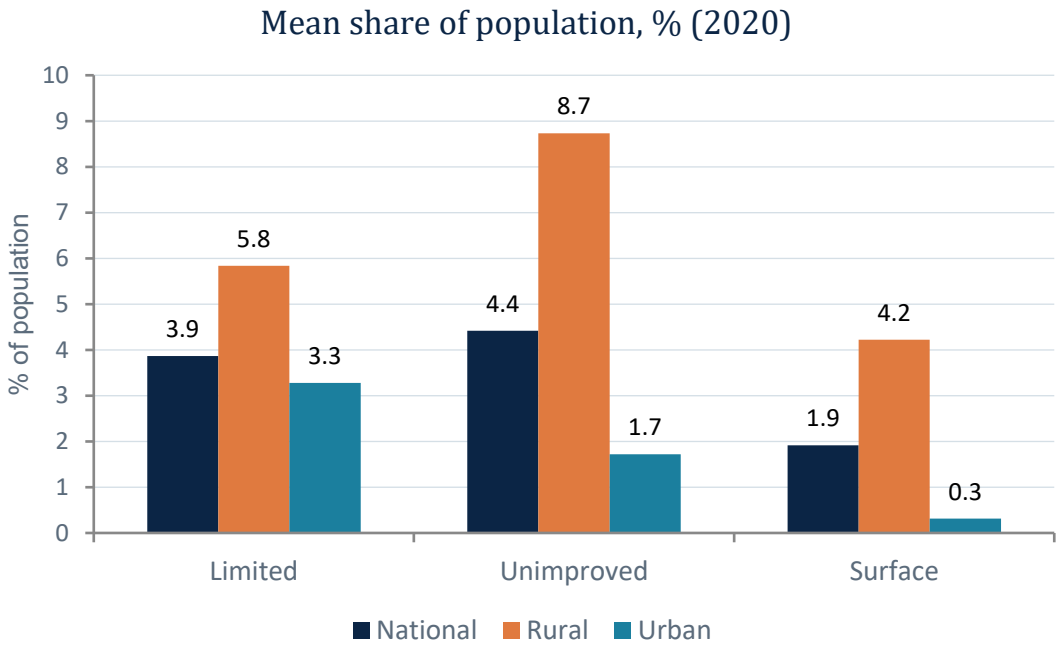


Basic %	Natl	Rural	Urban
Mean	89.9	81.3	94.7
Median	97.3	90.7	98.1
IQR	14.2	34.3	7.4
Lowest	37.2	22.0	49.7

Rural spread (IQR) is 4.6x the urban spread.

Most countries sit at or near 100% (mode = 100), pulling the median above the mean.

Where basic water is missing, rural people fall to the lowest rungs

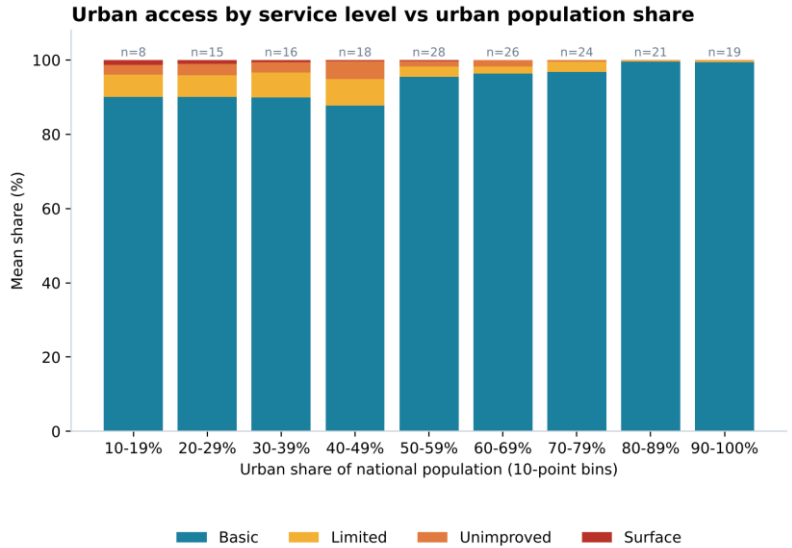


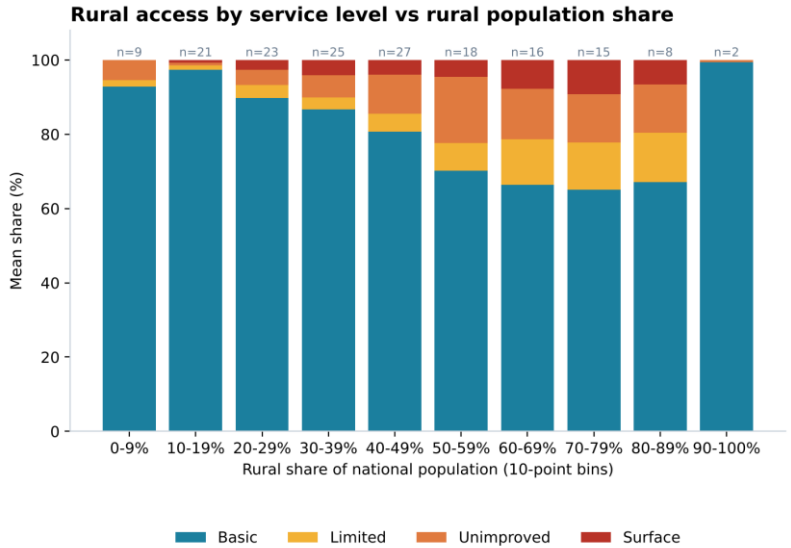
13.5x
rural vs urban reliance on surface water

5.1x
rural vs urban reliance on unimproved sources

94.7%
urban basic access vs 81.3% rural

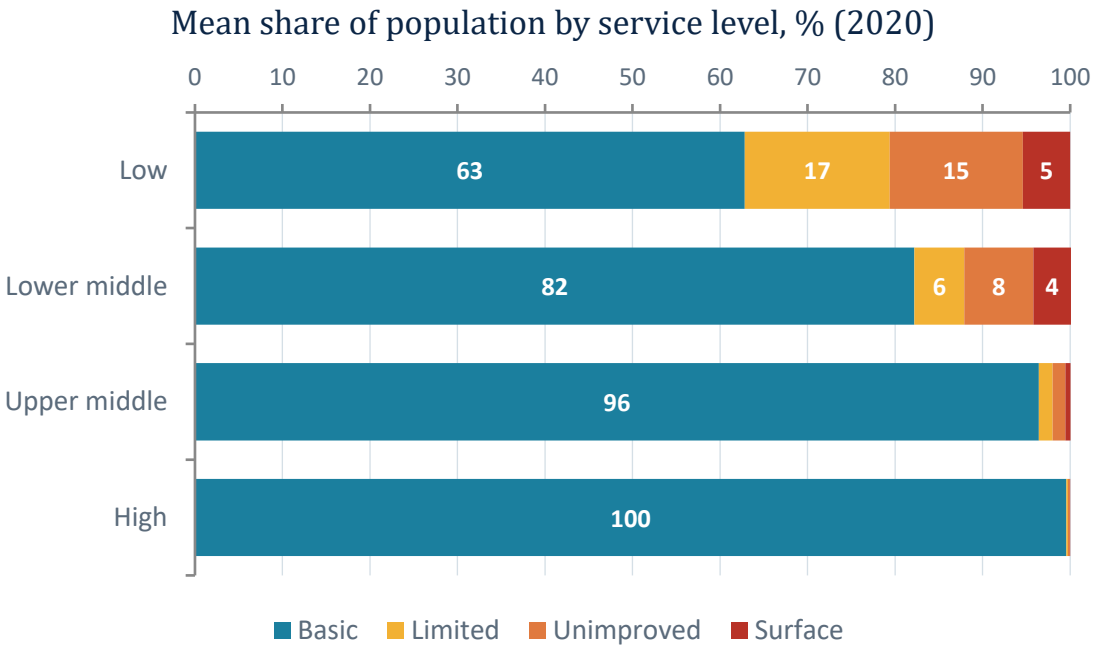
Access tracks how rural a country is, not how many people live there





Population size vs basic access: r = 0.00. Urban share vs basic access: r = 0.55.
Rural basic access is weakest (about 65-70%) in countries where 50-89% of people live in rural areas.

Income is the sharpest divider: low-income countries lag by 37 points



62.8%

basic access, low income (high income: 99.6%)

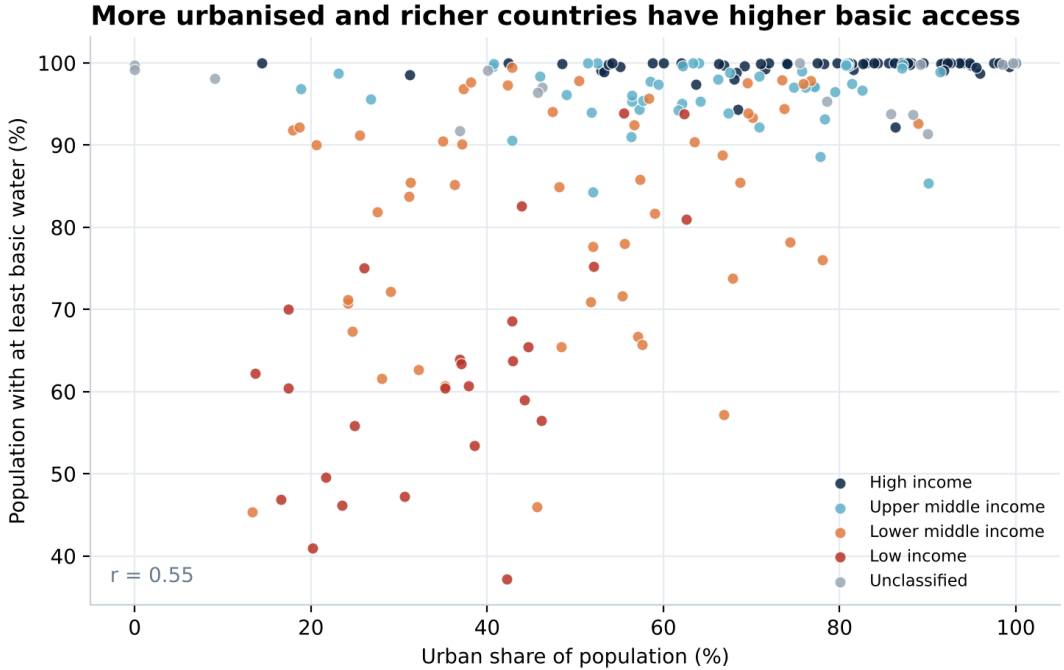
5.4%

rely on surface water (high income: 0.0%)

36% vs 79%

average urban share, low vs high income

Urbanisation and income together separate the countries

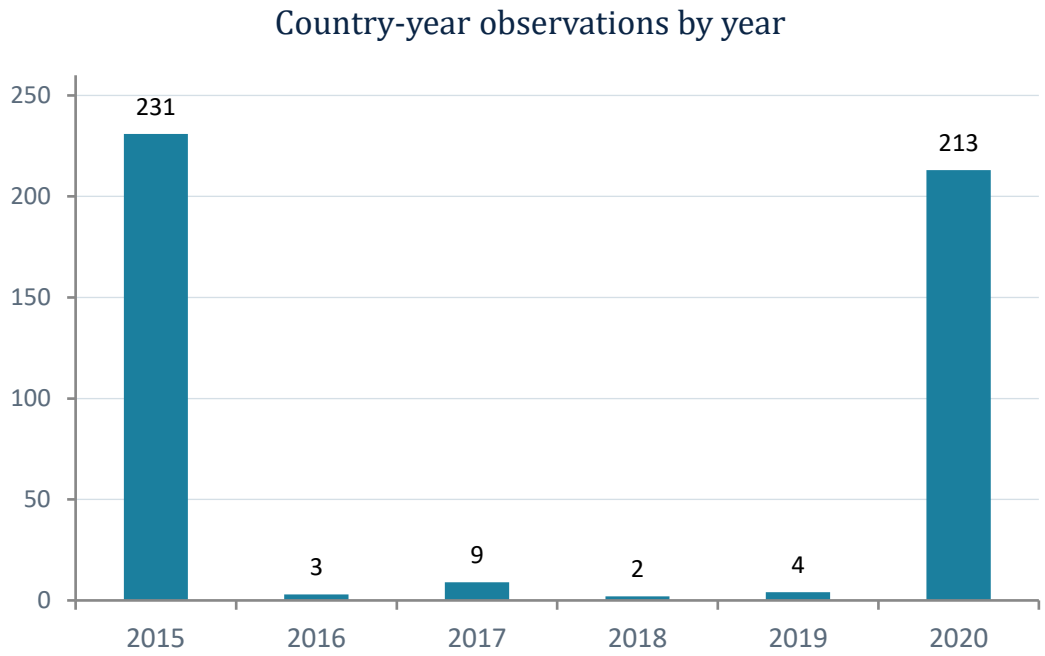


r = 0.55

Moderate positive link between urban share and basic access.

56 countries are below 90% basic access and 8 below 50%.

The Part two workbook file is really two snapshots: 2015 and 2020



4.8

average years between a country's two observations (range 1 to 5)

213 of 231

countries have the full 2015 to 2020 span

So change is measured per year, which keeps the 18 shorter spans comparable.

ARC turns two snapshots into a yearly rate of progress

```
ARC_n =
IFERROR( IF( $A3=$A2,
(E3-E2) / ($B3-$B2), ), "null" )
```

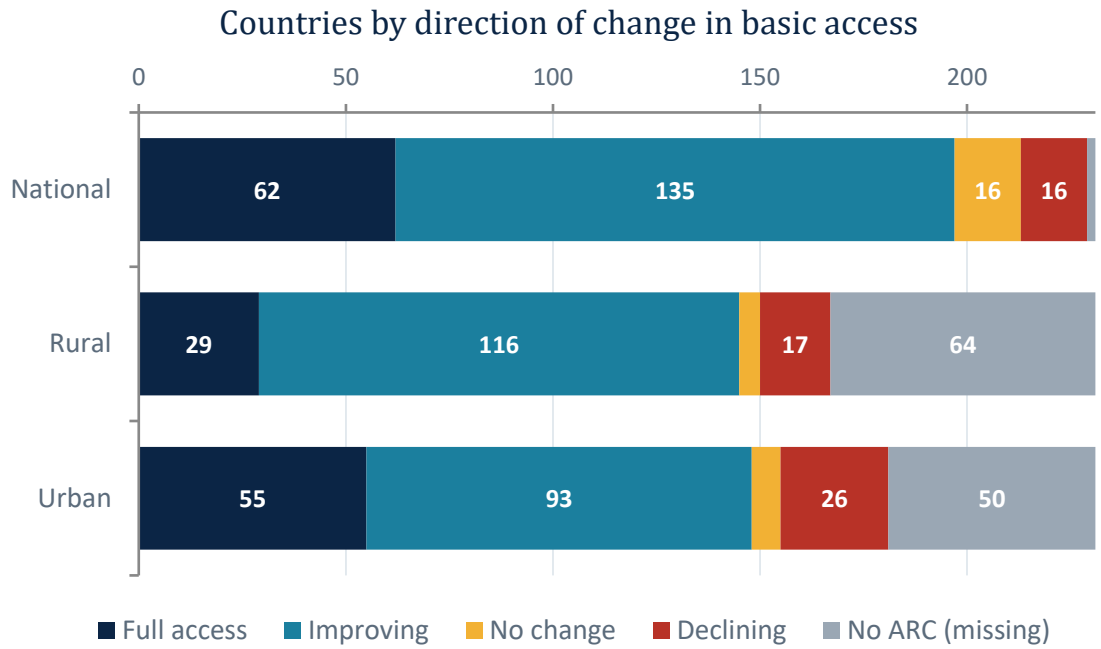
- Same-country test stops ARC crossing from one country to the next
- Mixed references ($A, relative row) copy one formula across n, r, u
- IFERROR labels missing estimates as null instead of #VALUE!
- Full access = rounded basic access of 100 in both years

ARC (pp/yr)	Avg	Min	Max
National	0.28	-1.02	2.75
Rural	0.48	-1.23	2.67
Urban	0.15	-1.62	2.67

0.48 vs 0.15

rural vs urban average ARC: rural access gained faster, from a lower start

Access improved in 135 of 231 countries, but rural gaps in the data limit the story



135 / 16 / 16

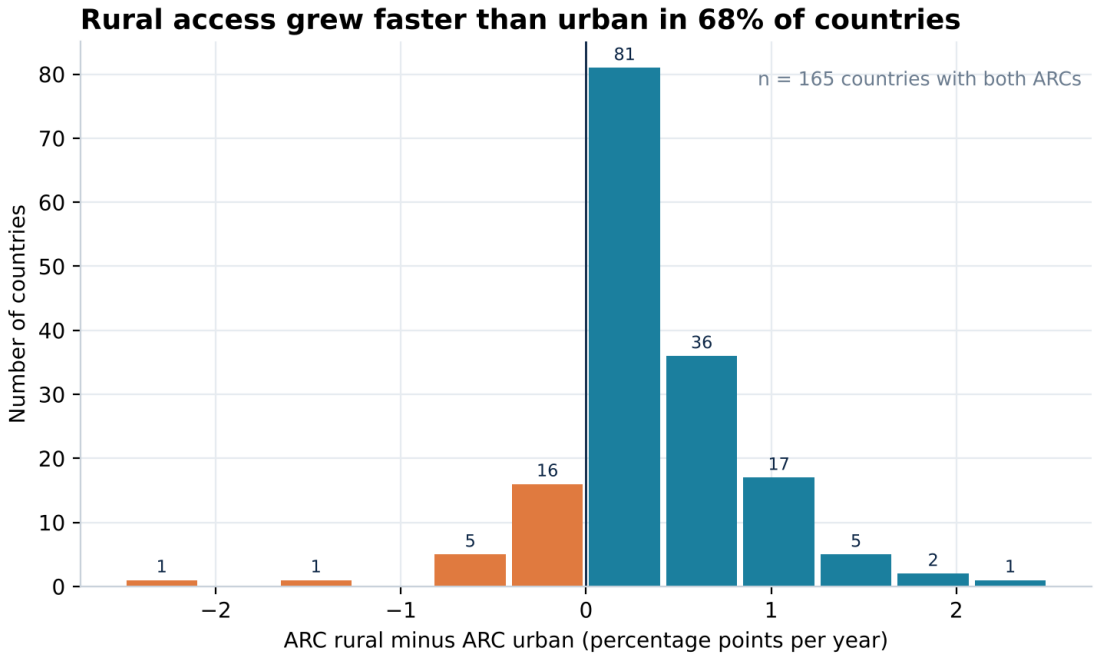
countries improving, stalled, declining nationally (not full access)

64 and 50

countries with no rural or urban ARC because an estimate is missing

Rural average ARC (0.48) beats urban (0.15) partly because urban areas start closer to 100%.

Rural access grew faster than urban in two out of three countries



68%

of 165 countries: rural ARC above urban (median gap 0.21 pp/yr)

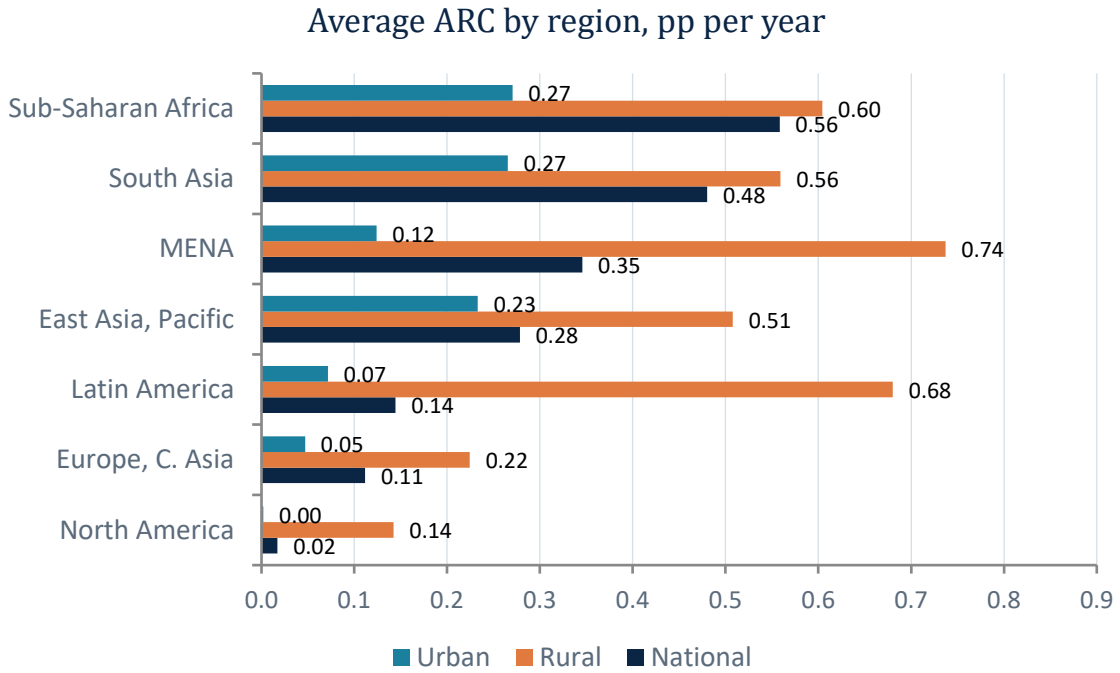
Largest gaps

Morocco: rural +2.63 vs urban +0.31

South Sudan: rural -0.63 vs urban +1.86

(pp per year)

Sub-Saharan Africa is improving fastest, from the lowest base



0.56

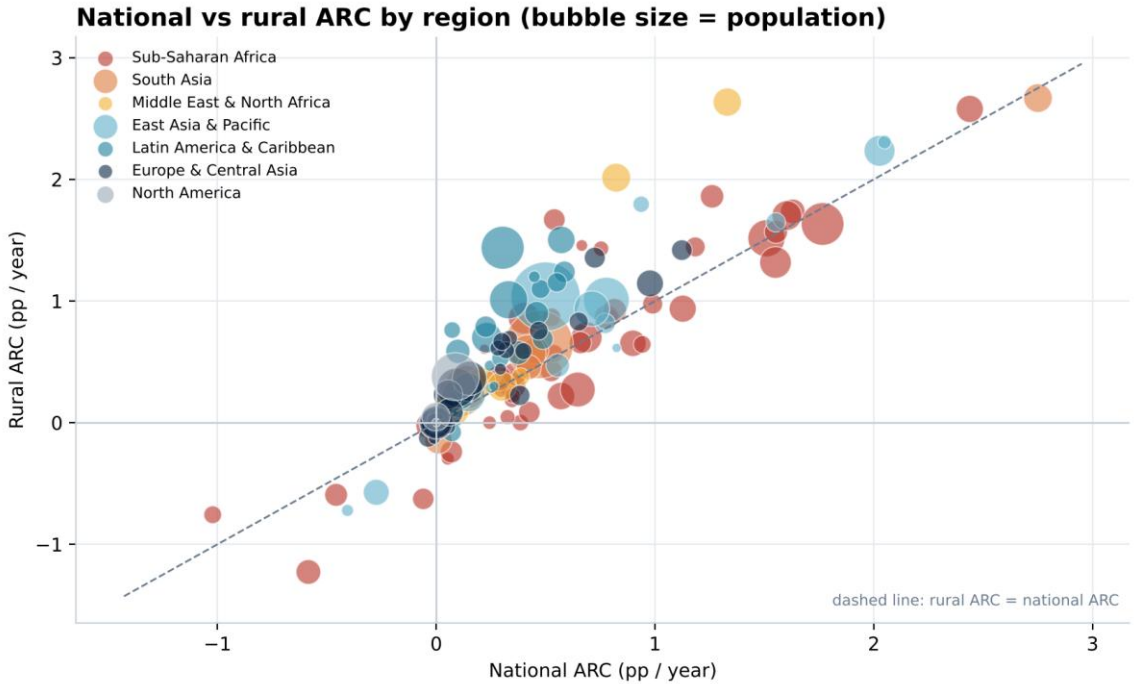
pp a year: highest national ARC, in Sub-Saharan Africa

37 of 44

countries below 80% basic access are in Sub-Saharan Africa

Europe and North America move slowly because most countries are already near 100%.

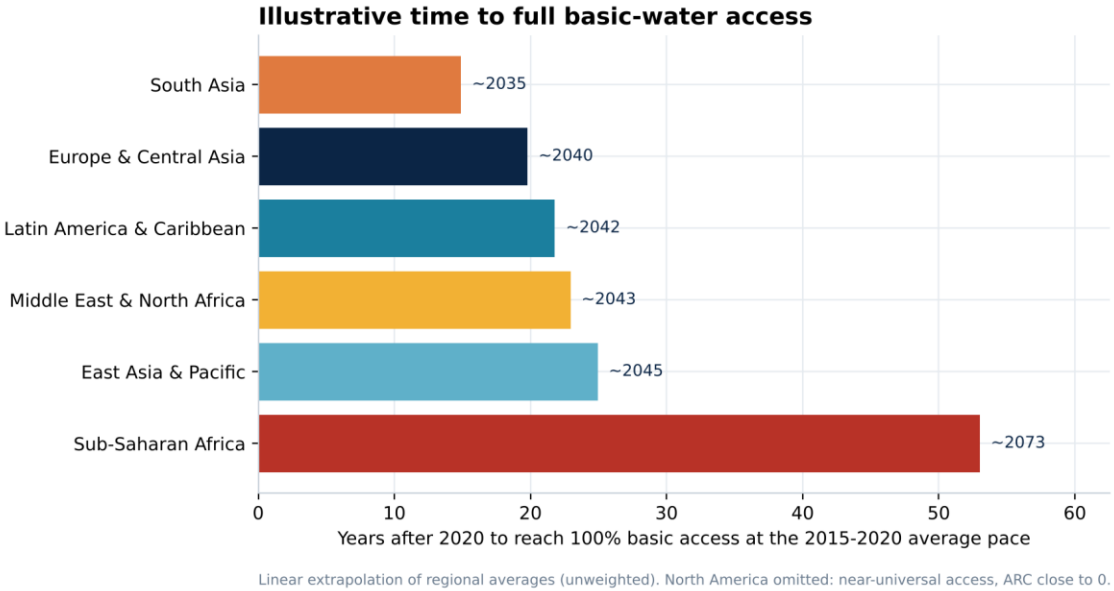
Sub-Saharan Africa has the widest range of national ARC



Reading the chart

- 109 of 167 countries sit above the line: rural ARC beat the national ARC
- Sub-Saharan Africa spans -1.0 to +2.4 pp a year nationally
- 11 of the 13 countries with over 100 million people sit between 0 and 1 pp a year

At the 2015-2020 pace, Sub-Saharan Africa needs roughly 50 more years



How to read it

Years to 100% = (100 minus average basic access) divided by average national ARC.

Caveat

Linear and unweighted. A signal about relative urgency, not a prediction.

What the data says, and what I would do next

1

Rural and low-income populations carry the gap

A 13-point rural deficit and a 37-point income deficit in basic access.

2

Progress is real but uneven

135 countries improved; 32 stalled or declined; Sub-Saharan Africa gains fastest but starts lowest.

3

Targeting beats averaging

Aim investment at rural areas of low-income, low-urbanisation, mostly Sub-Saharan countries.

4

Next: weight, extend, deepen

Population-weight the averages, add the safely managed level, and use more than two years.

APPENDIX

The 19 features I engineered

Part	Feature	What it does
Part 1	`value_cnt`	COUNTA of each row; should equal 16
Part 1	`pop_u_val, pop_r`	Urban head count (pop_n x pop_u / 100); rural share (100 - pop_u)
Part 1	`pop_n (m), wat_bas_n (rounded)`	Population rounded up to millions; basic access capped at 100%
Part 1	`pop_u (rounded), pop_r (rounded)`	Shares rounded to whole numbers for charting
Part 2	`y_diff`	Years between a country's two observations
Part 2	`ARC_n, ARC_r, ARC_u`	Annual rate of change in basic access (pp per year)
Part 2	`wat_bas_x (rounded) x3, ARC_x_full x3`	Rounded basic access and full-access flags
Part 2	`ARC_diff, region`	Rural minus urban ARC; region by lookup on country name